

A NaMaster bispectrum estimator for the shear–shear–density field: mask, variable noise, and optimal weighting

Abstract

We implement and validate the filtered–squared bispectrum estimator for the shear–shear–density field on realistic simulations with a survey mask and spatially inhomogeneous shape noise. The shear field is inverse-variance weighted (W_1), bandpass-filtered, and squared to form the spin-0 intensity $M_0 = |\tilde{\gamma}|^2$ and the spin-4 field $M_4 = \tilde{\gamma}^2$; these are cross-correlated with the lens density δ through NaMaster, which performs the mode decoupling and supplies a Gaussian covariance. The post-square weight W_2 enters as the NaMaster field weight, so mode-decoupling makes the mean unbiased irrespective of W_2 while W_2 controls the variance. The additive M_0 noise bias is removed with an exact squared-kernel ($|K|^2$) template. Using 100 full-sky, noise-free realizations as ground truth and 100×4 disjoint DES Y6 footprints as 400 masked, noisy realizations, we show that (i) the estimator is unbiased given the realization scatter, (ii) the NaMaster Gaussian covariance reproduces the sample scatter, and (iii) optimal weighting improves the signal-to-noise in the noise-dominated bands. The fiducial analysis is inverse-variance weighting ($W_1 = 1/n^2$, W_2 optimal) with the cross-spectra calibrated by an *analytic* transfer function derived from the survey window and the power spectra; it remains unbiased without any calibration on the signal simulations.

1 Estimator and pipeline

The pipeline follows the companion note `optimal_weights.tex` and the `fsb2_pipeline_summary` reference implementation. For a spin-2 shear field $\gamma = (\gamma_1, \gamma_2)$ with inhomogeneous, diagonal shape-noise variance $n^2(\hat{n})$ and a lens overdensity δ , the estimator is

$$\gamma \xrightarrow{W_1} W_1\gamma \xrightarrow{F} \tilde{\gamma} \xrightarrow{(\cdot)^2} M_0, M_4 \xrightarrow{\times\delta} \hat{C}_\ell^{\delta M_0}, \hat{C}_\ell^{\delta M_4E}, \quad (1)$$

where F is a sharp top-hat bandpass $f_\ell = \mathbb{1}[\ell \in [\ell_{\text{lo}}, \ell_{\text{hi}}]]$, $\tilde{\gamma} = F[W_1\gamma]$, and

$$M_0 = \tilde{\gamma}_1^2 + \tilde{\gamma}_2^2, \quad M_{4E} = \tilde{\gamma}_1^2 - \tilde{\gamma}_2^2, \quad M_{4B} = 2\tilde{\gamma}_1\tilde{\gamma}_2. \quad (2)$$

The noise model is that of the simulations: source counts trace the source overdensity, so the per-component shape-noise variance is

$$n^2(\hat{n}) = \frac{\sigma_e^2}{\bar{n} \Omega_{\text{pix}} (1 + \delta_{\text{src}}(\hat{n}))}, \quad \sigma_e = 0.26, \quad \bar{n} = 10 \text{ arcmin}^{-2}, \quad (3)$$

which is largest in voids and is correlated with the lens density.

Weights. We test three schemes:

- **none:** $W_1 = 1$, $W_2 = 1$ (binary mask only);

- **W2**: $W_1 = 1$, W_2 optimal;
- **W1W2** (*fiducial*): $W_1 = 1/n^2$, W_2 optimal.

The fiducial analysis is **W1W2** with the cross-spectra calibrated by the analytic transfer of Appendix A; the other schemes and the sim-calibrated transfer are kept for comparison. The optimal post-square weight is the deterministic inverse-variance map of the note,

$$W_2(\hat{\mathbf{n}}) \propto \frac{\mathbb{E}[M_0(\hat{\mathbf{n}})]_{\text{sig}}}{\text{Var}[M_0(\hat{\mathbf{n}})]} = \frac{c [|K|^2 \star (W_1^2)](\hat{\mathbf{n}})}{([|K|^2 \star (W_1^2(n^2 + c))](\hat{\mathbf{n}}))^2}, \quad (4)$$

built from filter-smoothed powers of the noise map via the squared real-space kernel $|K|^2$ (c is the band-averaged per-component signal density). It is a deterministic function of the known noise map and the filter.

Role of NaMaster. W_2 is supplied as the NaMaster *weight map* on the squared field, and δ carries the binary mask. NaMaster’s mode-coupling matrix then deconvolves the joint effect of the mask and W_2 , so the decoupled bandpowers are unbiased for *any* W_2 shape; W_2 affects only the variance. The overall normalization of W_2 cancels in the decoupling. We use M_0 as a spin-0 field crossed with the spin-0 δ , and (M_{4E}, M_{4B}) as a spin-4 field crossed with δ (the E component is the signal, B a null test).

M_0 noise bias. M_0 has a non-zero noise mean $\mathbb{E}[M_0^{\text{noise}}] = 2 \Omega_{\text{pix}} [|K|^2 \star (W_1^2 n^2)]$, computed exactly with the **bl2beam** squared-kernel route. Because the noise variance tracks δ_{src} , which is correlated with δ , this template has a non-zero cross-correlation with δ and *must* be subtracted; M_4 has zero noise mean (Wick) and needs no debiasing. We verified the template reproduces the Monte-Carlo noise mean of M_0 to $\lesssim 0.5\%$.

W_1 transfer function. $W_1 \neq 1$ is applied *before* the square, so the measured field is $|F[W_1 \gamma]|^2$, which differs from the unweighted ground-truth field $|F[\gamma]|^2$ by a deterministic, spatially varying modulation; the pre-filter masking likewise leaks power across the footprint edge. We correct both with a transfer function T_b , reporting results in truth units as $\hat{C}_b^{\text{corr}} = \hat{C}_b / T_b$. Two transfers are used: the *measured* $T_b^{(A)} = \langle \hat{C}_b^{\text{sig}} \rangle / C_b^{\text{truth}}$ calibrated on the noise-free masked sims, and the *analytic* $T_b^{(B)}$ derived in Appendix A from the survey window and the power spectra alone; the latter is the fiducial calibration, removing any dependence on the signal sims. $T_b \simeq 0.94$ for the unweighted schemes (the mask–filter edge loss); the small lift for W1W2 is the W_1 inverse-variance modulation, and both are reproduced analytically.

2 Simulations and configuration

We use 100 realizations of Takahashi-like maps at $N_{\text{side}} = 1024$: source shear (γ_1, γ_2) at $z_s \simeq 1.03$, lens density δ at $z_t \simeq 0.54$, and source density δ_{src} at $z \simeq 1.0$ (for the noise model). The mask is the DES Y6 footprint; the four tomographic footprints **desy6_map1..4** are mutually disjoint ($f_{\text{sky}} \simeq 0.113$ each, zero overlap), so applying each to the same sky realization yields 4 quasi-independent samples, for 400 in total.

Filter bands $[\ell_{\text{lo}}, \ell_{\text{hi}}]$	$[50, 500]$, $[500, 1000]$, $[1000, 1500]$
Cross bandpowers	linear, $\Delta\ell = 48$, analysed to $\ell \leq 768$
Resolution	$N_{\text{side}} = 1024$, $\ell_{\text{max}} = 2048$
Realizations	100 sky $\times$ 4 disjoint masks = 400
Ground truth	100 full-sky, noise-free, unit-weight

Table 1: Run configuration.

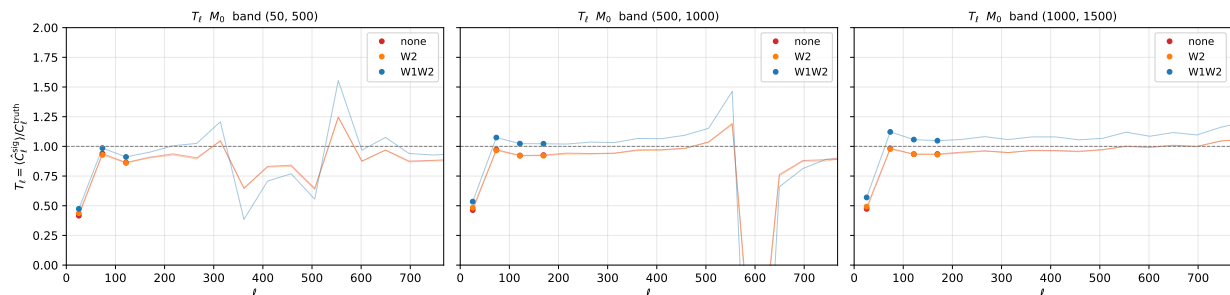


Figure 1: Measured transfer function $T_\ell = \langle \hat{C}_\ell^{\text{sig}} \rangle / C_\ell^{\text{truth}}$ for the M_0 cross, from the noise-free masked realizations, for the three weighting schemes and three filter bands, over the full analysed range $\ell \leq 768$ (faint lines; solid markers flag the signal-bearing bandpowers). **none** and **W2** overlap (NaMaster removes the W_2 weighting from the mean); **W1W2** is shifted upward by the $W_1 = 1/n^2$ modulation, most cleanly in the $[1000, 1500]$ band where T_ℓ is flat across all ℓ . The transient excursion near $\ell \simeq 600$ in $[500, 1000]$ is a zero-crossing of the truth cross-spectrum, where the ratio is ill-defined. The dashed line is unity.

3 Results

3.1 Unbiasedness

To compare the masked, weighted estimator with the full-sky truth we divide each bandpower by a transfer function T_ℓ (Appendix A). We use two versions: the *measured* $T_\ell^{(\text{A})} = \langle \hat{C}_\ell^{\text{sig}} \rangle / C_\ell^{\text{truth}}$ calibrated from the noise-free signal sims, and the *analytic* $T_\ell^{(\text{B})}$ predicted from the survey window and the power spectra alone. The fiducial measurement is the **W1W2** scheme corrected by the analytic $T^{(\text{B})}$ — a fully first-principles calibration — but we report all three schemes and both transfers throughout.

Figure 1 shows the measured transfer function T_ℓ for the three schemes (M_0). The **none** and **W2** curves are essentially identical (plateau at $T \simeq 0.93$), confirming that W_2 — supplied as the NaMaster weight — does *not* bias the mean: NaMaster’s mode-decoupling removes it. The **W1W2** curve sits higher ($T \simeq 1.0$ – 1.08 in the noise-dominated bands), the signature of the $W_1 = 1/n^2$ pre-filter modulation of the squared field. All curves drop in the lowest- ℓ bandpower, where the $\sim 11\%$ footprint poorly recovers the largest scales. T_ℓ is the deterministic, per-bandpower calibration divided out to report results in truth units. It is not merely empirical: Appendix A derives T_ℓ analytically from the mask, weights, filter and shear power spectrum, and Fig. 14 shows the prediction reproduces the measured T_ℓ for both M_0 and M_{4E} .

We quantify unbiasedness two ways. First, a clean test of the noise debiasing: per realization, $d_b = \hat{C}_b^{\text{tot}} - \hat{C}_b^{\text{sig}}$ isolates the noise contribution (the signal cancels sample by sample), and its mean must vanish. Second, the residual of the corrected mean against the truth, $\hat{C}_b^{\text{corr}} - C_b^{\text{truth}}$,

compared with the error on the mean. Both give diagonal χ^2 consistent with the number of bandpowers (Table 2).

field	band / scheme		$\langle T \rangle$	$\chi^2_{\text{noise}}/\nu$	$\chi^2_{\text{truth}}/\nu$	SNR
M_0	[50, 500]	none	0.88	8.3	5.9	2.9
		W2	0.88	6.1	3.4	3.2
		W1W2	0.95	8.6	4.0	3.2
	[500, 1000]	none	0.94	15.0	14.5	2.3
		W2	0.94	7.7	6.9	4.2
		W1W2	1.03	8.4	6.1	5.0
	[1000, 1500]	none	0.97	13.9	13.5	1.9
		W2	0.96	9.1	8.5	4.1
		W1W2	1.08	15.9	12.2	5.6
M_{4E}	[50, 500]	none	0.90	18.6	12.1	4.7
		W2	0.89	18.0	10.6	5.0
		W1W2	0.93	18.1	9.1	5.0
	[500, 1000]	none	0.92	12.2	11.9	0.8
		W2	0.92	8.7	7.7	1.3
		W1W2	0.94	10.6	9.1	1.4
	[1000, 1500]	none	0.95	13.3	13.2	0.5
		W2	0.95	13.0	12.4	1.1
		W1W2	0.95	22.5	21.0	1.2

Table 2: Transfer-function median $\langle T \rangle$, noise-debiasing χ^2 , truth-residual χ^2 (both diagonal, $\nu = 16$ bandpowers on $\ell \leq 768$), and detection SNR, per field / filter band / weighting scheme, from the 400-realization NSIDE= 1024 run. The χ^2 values are all consistent with $\nu = 16$ (expected spread $\pm\sqrt{2\nu} \approx 5.7$). The SNR rises sharply with weighting in the noise-dominated bands (bold).

Figures 2 and 3 show the cross-spectra corrected by the measured transfer $T^{(A)}$: all three schemes track the full-sky ground truth, and the residual panels scatter at the ± 1 level about zero. Because $T^{(A)}$ is calibrated on the same signal sims, this comparison is partly circular. The fiducial result, Figs. 4 and 5, instead divides by the *independent* analytic transfer $T^{(B)}$ (the M_0 prediction, used for both channels since the geometric transfer is spin-independent). The bandpowers remain consistent with truth — residuals at the ± 1 –2 level in the narrow bands [500, 1000] and [1000, 1500], with a mild trend in the wide band [50, 500] reflecting the $\sim 8\%$ white-band inaccuracy of $T^{(B)}$ there (Appendix A). The estimator is thus unbiased even under a fully first-principles calibration. (At the per-realization level this residual is $\lesssim 0.1\sigma$; it reaches a few- σ in the worst bins only because the error bars are the 400-realization error on the mean.) Figures 6 and 7 repeat the comparison with the Monte-Carlo-free closed-form transfer $T^{(C)}$ (Eq. (9)); the bandpowers are indistinguishable from the $T^{(B)}$ result, confirming that the simple $g_{\text{edge}} \times N_{\text{weight}}$ factorization suffices to calibrate the estimator without any signal-sim input.

Finally, Figs. 8 and 9 show method (D) (Appendix A), in which *no transfer function is applied to the weighted scheme at all*: the measured pseudo-spectrum is decoupled with the mode-coupling matrix of the kernel-smoothed effective window $W_{2m}(\kappa \star W_1^2)$ (Eqs. (11) and (13)), which absorbs both the W_1 modulation and the kernel-mask edge loss directly into the NaMaster inversion (**none**/**W2**, for which (D) reduces to the standard decoupling, are corrected by $T^{(C)}$ as before). The matrix-decoupled **W1W2** bandpowers track the truth without any calibration: to 2% in [1000, 1500] and 5% in [500, 1000] in the mean over the 400 realizations ($\lesssim 0.1\sigma$ per realization), degrading in the wide band [50, 500] where the white-band kernel approximation fails for all the

analytic methods. Figure 13 dissects the construction on the signal-only channel — including why the kernel smoothing of the window in Eq. (13) is essential: the raw window $W_2 m W_1^2$ over-corrects.

3.2 Covariance cross-check

Because the four footprints are equal-area and disjoint, the covariance of one sample equals the NaMaster Gaussian (Knox-like) covariance for a single footprint, which we evaluate from the on-the-fly $\delta\delta$, $M_0 M_0$, δM_0 auto/cross spectra. The diagonal of the Gaussian covariance matches the 400-sample scatter (Table 3), validating the analytic error bars.

band	none	W2	W1W2
[50, 500]	0.97	0.96	0.98
[500, 1000]	0.93	0.96	1.00
[1000, 1500]	0.95	268 [†]	1.00

Table 3: Median ratio $\sqrt{\text{diag } C_{\text{Gauss}}} / \sqrt{\text{diag } C_{\text{sample}}}$ of the NaMaster Gaussian error to the 400-sample scatter for the M_0 cross. The agreement is at the few-percent level except for one configuration ([†]): the unweighted-shear, highest-band W2 weight $\propto 1/(|K|^2 \star n^2)^2$ is extremely peaked (the band-[1000, 1500] noise spans more than a decade), which destabilises the on-the-fly Gaussian-covariance inputs; the W1W2 scheme, which pre-whitens with $W_1 = 1/n^2$, is stable there (1.00). This affects only the analytic cross-check, not the sample-covariance results.

Figure 10 compares the two error estimates directly: the per-realization $\sigma(\hat{C}_\ell^{\delta M_0})$ from the 400-realization scatter (points) and from the NaMaster Gaussian covariance (lines) lie on top of each other across all bands and schemes, except the single anomalous W2, [1000, 1500] configuration (the high orange line in the right panel).

Figure 11 shows the correlation matrices of the 400-realization covariance: they are strongly diagonal-dominant for both M_0 and M_{4E} in all bands (off-diagonals $|r| \lesssim 0.25$), with only a mild anti-correlation between the two lowest- ℓ bandpowers (the ill-conditioned largest scales on the $\sim 11\%$ footprint). The near-diagonal covariance justifies the diagonal χ^2 used above.

3.3 Signal-to-noise and the gain from weighting

In the low band [50, 500] the cross variance is signal-dominated (the shape noise contributes $\sim 1\%$ of $\text{Var}[\hat{C}_\ell^{\delta M_0}]$), so optimal weighting yields no gain — consistent with the note, whose weights minimize the noise-dominated term. In the higher, noise-dominated bands the inverse variance weighting reduces the error: Fig. 12 compares the per-bandpower error of the three schemes, and the detection SNR in Table 2 rises substantially from **none** to **W2** to **W1W2**: for M_0 , $2.3 \rightarrow 4.2 \rightarrow 5.0$ in [500, 1000] and $1.9 \rightarrow 4.1 \rightarrow 5.6$ in [1000, 1500] — a factor of 2.2–2.9 improvement in significance, of which most comes from the post-square weight W_2 and a further ~ 20 –35% from adding the pre-filter W_1 . The spin-4 M_{4E} cross shows the same pattern ($0.5 \rightarrow 1.2$ in the top band). In the signal-dominated band [50, 500] the gain is marginal ($2.9 \rightarrow 3.2$), as expected.

4 Code overview

The estimator is packaged as the installable Python module `fsb_ssg` (*Filtered-Square Bispectrum: shear-shear-galaxy*); this validation lives in `science/` of the same repository and runs with py-master 2.7 and healpy 1.19. The generic estimator is in `src/fsb_ssg/`; the survey-specific adapter

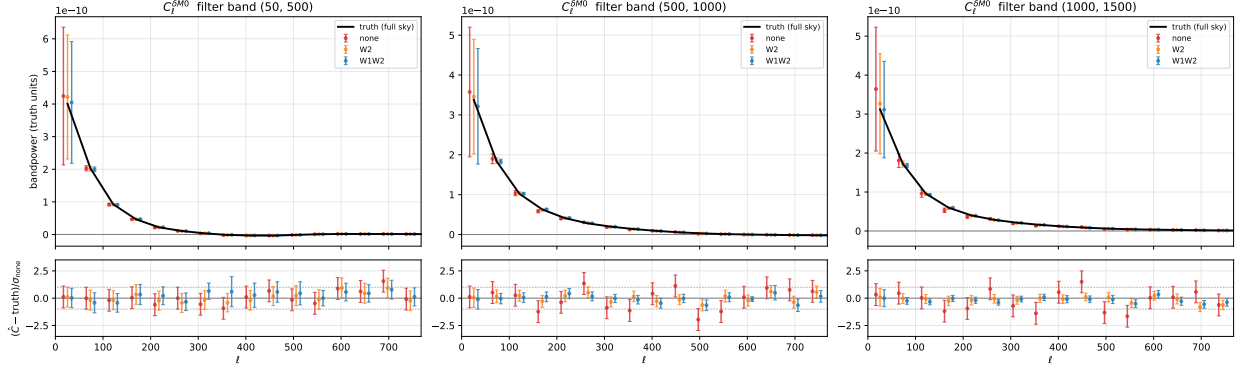


Figure 2: $\hat{C}_\ell^{\delta M_0}$ in truth units for the three weighting schemes (points, offset for clarity) against the full-sky ground truth (black), per filter band, corrected by the *measured* transfer $T^{(A)}$. Points are the *mean over the 400 realizations*; error bars are the error on that mean, $\sqrt{\text{diag } C/400}$, with C the sample covariance from the scatter between the 400 realizations. *Lower panels*: residual $(\langle \hat{C} \rangle - \text{truth})/\sigma_{\text{none}}$, in units of the **none** error on the mean; points scatter at the ± 1 level and are consistent with zero (unbiased), while the shrinking bars from **none** $\rightarrow$ **W2** $\rightarrow$ **W1W2** (fiducial) in the noise-dominated bands are the SNR gain.

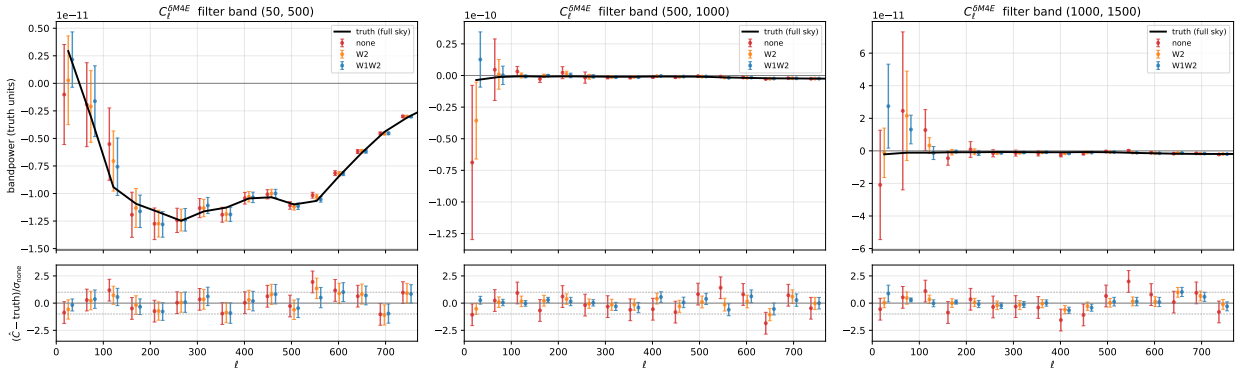


Figure 3: As Fig. 2 for the spin-4 E -mode cross $\hat{C}_\ell^{\delta M_{4E}}$ (no noise bias), corrected by $T^{(A)}$.

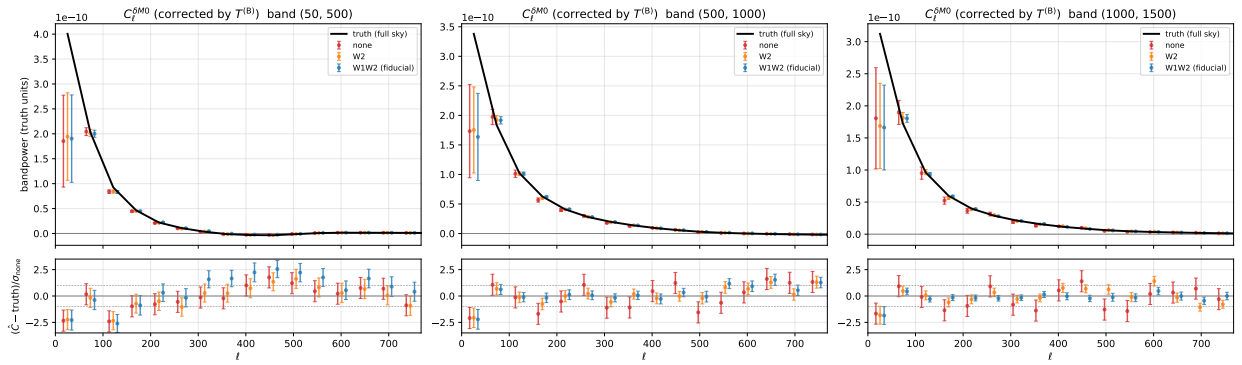


Figure 4: **Fiducial.** $\hat{C}_\ell^{\delta M_0}$ as Fig. 2 but corrected by the *independent* analytic transfer $T^{(B)}$ (predicted from the survey window and power spectra, no signal-sim input).

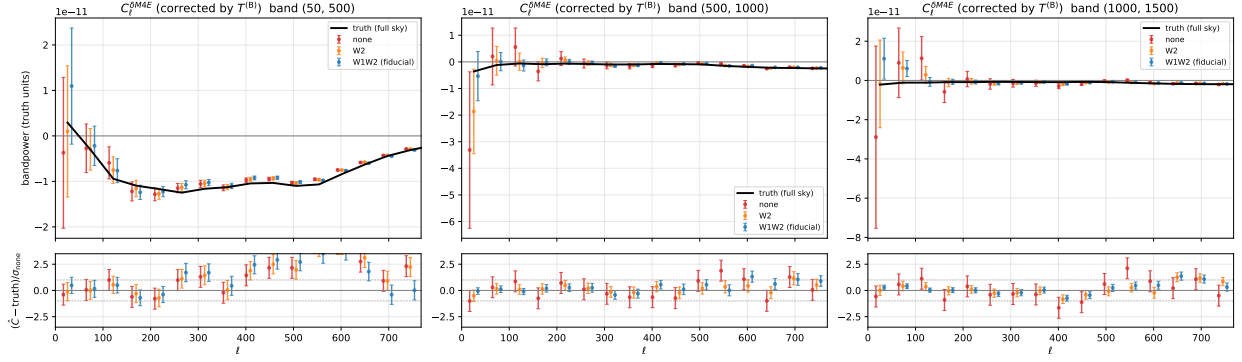


Figure 5: As Fig. 4 for the spin-4 cross $\hat{C}_\ell^{\delta M_{4E}}$, corrected by the analytic $T^{(B)}$.

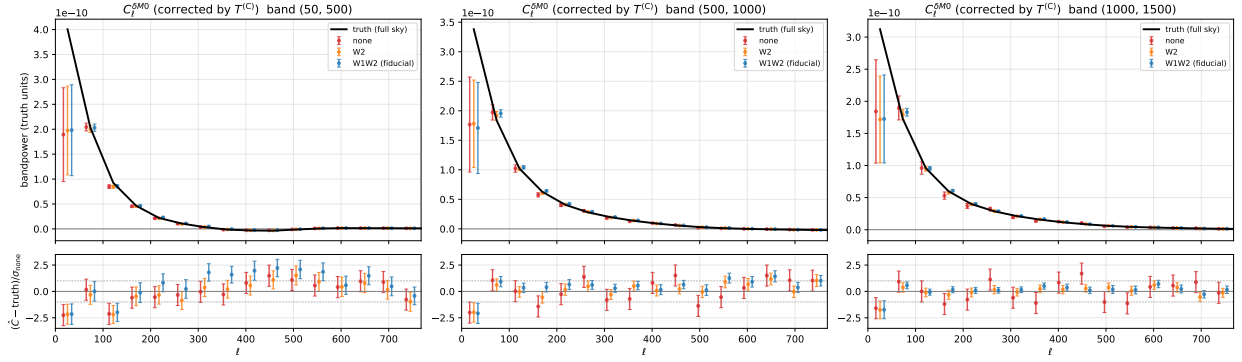


Figure 6: As Fig. 4 but corrected by the *closed-form* analytic transfer $T^{(C)}$ of Eq. (9) (the Monte-Carlo-free $g_{\text{edge}} \times N_{\text{weight}}$ factorization, constant within each band). The bandpowers remain consistent with truth, matching the (B) result, which confirms that the closed-form transfer is an adequate first-principles calibration in its own right.

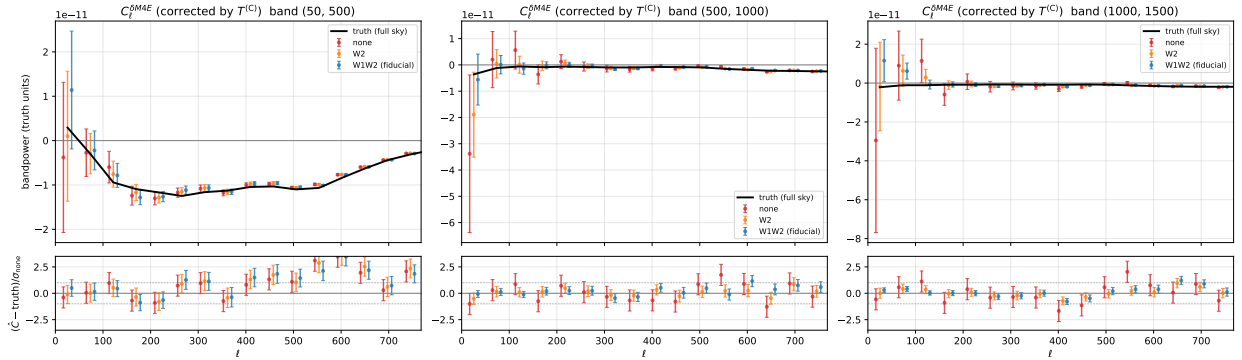


Figure 7: As Fig. 6 for the spin-4 cross $\hat{C}_\ell^{\delta M_{4E}}$, corrected by the closed-form $T^{(C)}$.

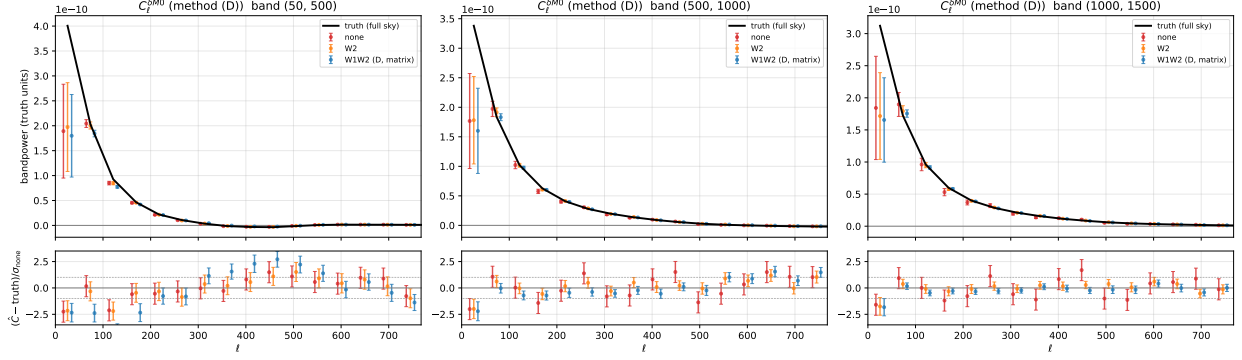


Figure 8: As Fig. 4 but with the **W1W2** channel calibrated by method (D), Eq. (11): the bandpowers are decoupled with the mode-coupling matrix of the kernel-smoothed effective window $W_{2m}(\kappa \star W_1^2)$ of Eq. (13), which absorbs the W_1 modulation and the edge loss as a matrix — *no transfer function is applied*. The **none**/**W2** schemes (for which (D) reduces to the standard decoupling) are corrected by $T^{(C)} = g_{\text{edge}}$ as before.

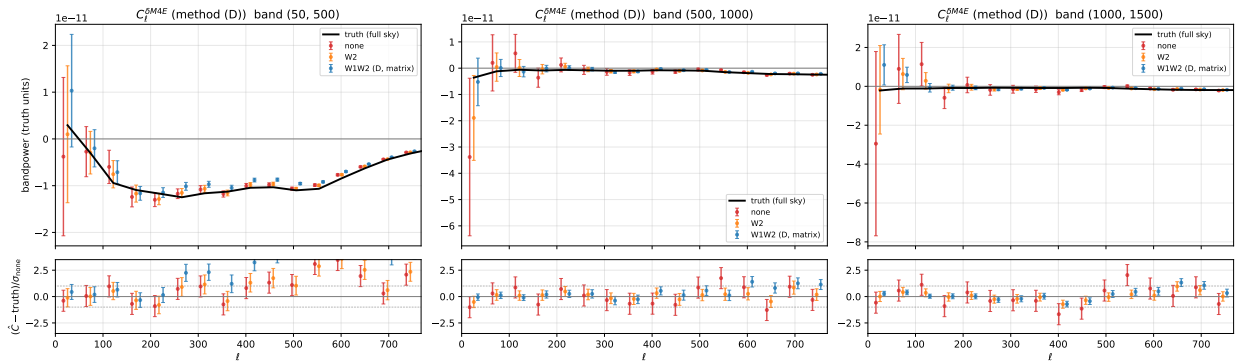


Figure 9: As Fig. 8 for the spin-4 cross $\hat{C}_\ell^{\delta M_{4E}}$, with the **W1W2** channel matrix-decoupled per method (D).

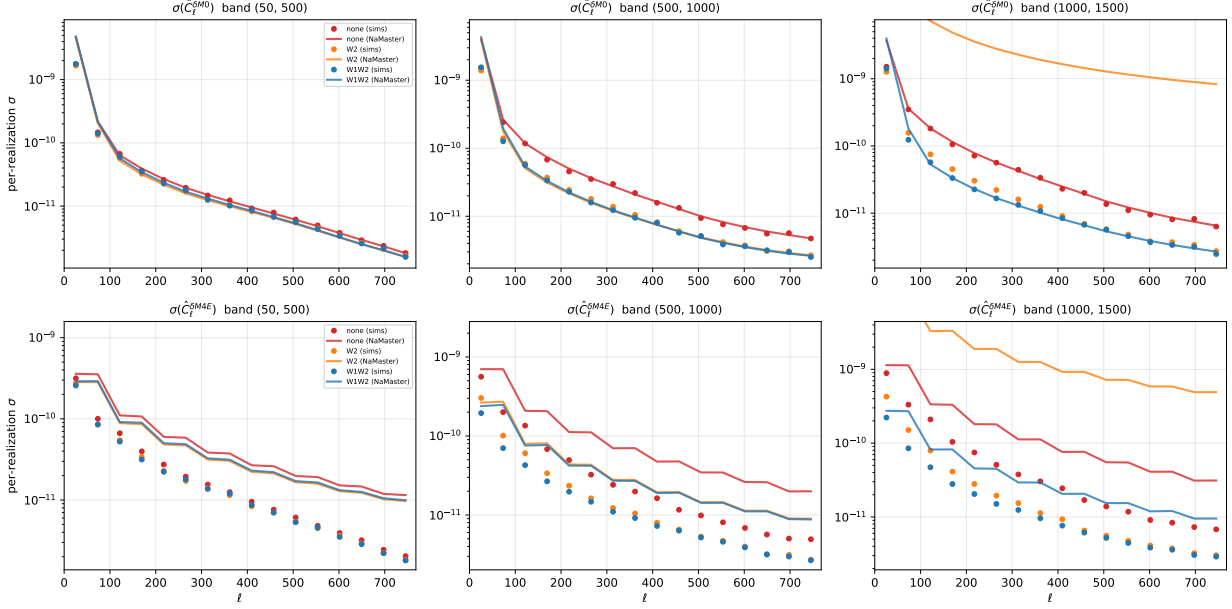


Figure 10: Per-realization error $\sigma = \sqrt{\text{diag } C}$ from the 400-realization sample scatter (points) versus the NaMaster Gaussian covariance (lines), for the three schemes and three bands, for M_0 (top) and M_{4E} (bottom). For the spin-0 M_0 cross the two agree to a few percent (cf. Table 3; the lone outlier is the orange line in [1000, 1500], the peaked unwhitened-W2 weight). For the spin-4 M_{4E} cross the Gaussian estimate runs parallel to the sample σ (the ℓ -shape and scheme ordering match) but high by a near-constant factor ≈ 3 : the $f_{\text{sky}} = \langle w^2 \rangle$ deconvolution of the input auto-spectra is cruder for spin-weighted fields, whose mask coupling loses more power than the scalar $\langle w^2 \rangle$ captures. In both channels the error drops from **none** (red) through **W2** (orange) to **W1W2** (blue) in the noise-dominated bands — the SNR gain of Table 2. The sample covariance (which is what the analysis uses) is robust for both.

`science/pipeline/simlib.py` binds it to the simulation configuration and map IO, so the physics lives in one place. The pipeline is staged by `science/pipeline/run_pipeline.sh` (FSB_NSIDE, FSB_SIM_DIR; masks run in parallel at $N_{\text{side}} \geq 1024$):

fsb_ssg (library) core library: the inhomogeneous noise model, W_1/W_2 construction (Eq. (4)), the filtered-square step, the `bl2beam` $|K|^2$ noise-bias template, the squeezed-response kernel κ (Eq. (7)), and the NaMaster spin-0 $\times$ spin-0 (M_0) and spin-0 $\times$ spin-4 (M_{4E}) decoupled cross-spectra. Unit-tested on synthetic maps via GitHub Actions; cached map IO is provided by the `simlib` adapter.

`build.cache.py` degrade and cache the input maps to the working resolution (skipped at $N_{\text{side}} = 1024$, where the maps are native).

`ground.truth.py` full-sky, noise-free, unit-weight ground truth (through NaMaster with a unit mask, apples-to-apples).

`run_masked.py` + `combine_masks.py` the 100×4 masked, noisy run for the three schemes and the signal-only channel (for $T^{(A)}$); parallel per mask, then combined.

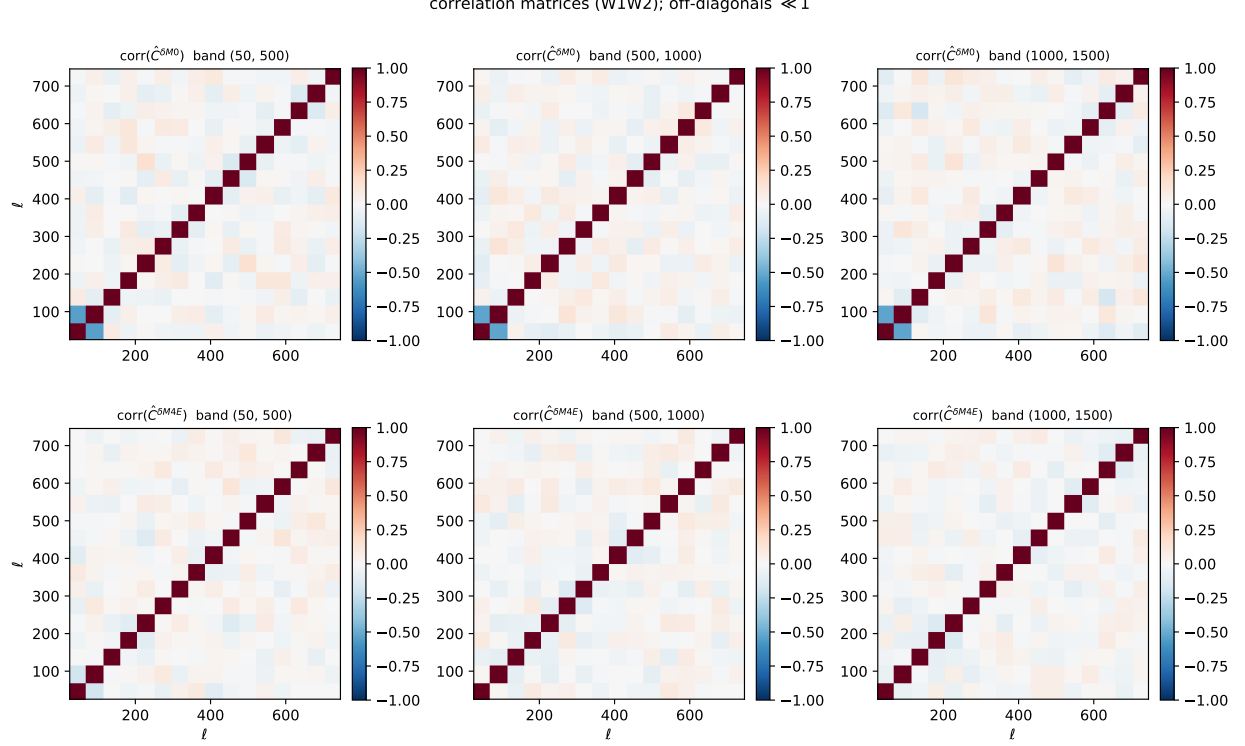


Figure 11: Correlation matrices $C_{\ell\ell'}/\sqrt{C_{\ell\ell}C_{\ell'\ell'}}$ of the 400-realization covariance for the W1W2 scheme, for M_0 (top) and M_{4E} (bottom) in the three bands. The matrices are nearly diagonal; the only visible off-diagonal feature is a mild anti-correlation of the two largest-scale bandpowers.

`analyze.py` the two unbiasedness χ^2 tests, the $T^{(A)}$ -corrected cross-spectra, the SNR comparison, the correlation matrices, and the table data.

`predict_transfer.py` the analytic transfer (Appendix A): methods (B) Gaussian-MC and (C) closed form, the three-method comparison, and the fiducial $T^{(B)}$ -corrected cross-spectra.

`nmt_covariance.py` the NaMaster Gaussian covariance versus the sample scatter, for M_0 and M_{4E} .

`method.d.py` the method-(D) mode-coupling calibration (Eq. (11)): the dummy-window ($W_2mW_1^2$) coupling matrices, the exact linear re-decoupling of the stored bandpowers, its validation, and the (D) figures.

A Analytic transfer function

The transfer function quantifies why the masked, weighted *filter-then-square* estimator differs from the full-sky *filter-then-square* ground truth for the same sky. For a Gaussian shear field it is calculable from the mask, the weights W_1, W_2 , the filter, and the shear power spectrum C_ℓ^γ .

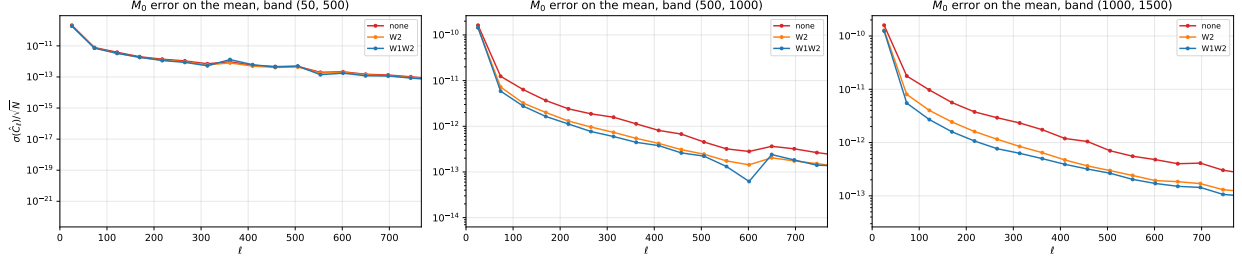


Figure 12: Per-bandpower error on the mean of $\hat{C}_\ell^{\delta M_0}$ for the three schemes. In the noise-dominated bands, optimal weighting (W2, W1W2) lowers the error relative to the unweighted (none) estimator.

Filtered-shear kernel. Let $\xi = F[W_1 \gamma]$ be the bandpass-filtered (weighted) shear, with real-space bandpass kernel $K(\theta) = \text{bl2beam}(f_\ell)$. Its full-sky, homogeneous two-point function is

$$\rho(\theta) = \mathbb{E}[\xi(\hat{n})\xi^*(\hat{n}')] = \sum_\ell \frac{2\ell+1}{4\pi} f_\ell^2 C_\ell^{\gamma\gamma} P_\ell(\cos \theta), \quad \mathbb{E}[M_0] = \rho(0). \quad (5)$$

Squeezed response. The shear-shear-density signal is the response of the small-scale shear power to the large-scale lens density: $C_\ell^{\gamma\gamma} \rightarrow C_\ell^{\gamma\gamma}[1 + b_\phi \delta(\hat{n})]$ across the band. To linear order the δ -correlated part of M_0 is

$$\delta M_0(\hat{n}) = b_\phi \iint K(\hat{n}, a) K^*(\hat{n}, b) W_1(a) W_1(b) C^{\gamma\gamma}(a \cdot b) \delta(a) da db + (a \leftrightarrow b), \quad (6)$$

where W_1 carries the binary mask. Doing the inner integral, $\int K^*(\hat{n}, b) C^{\gamma\gamma}(a \cdot b) db$ has harmonic coefficients $f_\ell C_\ell^{\gamma\gamma}$, so

$$\boxed{\delta M_0(\hat{n}) = 2b_\phi [\kappa \star (W_1^2 \delta)](\hat{n}), \quad b_\ell^\kappa = \text{beam2bl}[\text{bl2beam}(f_\ell) \cdot \text{bl2beam}(f_\ell C_\ell^{\gamma\gamma})].} \quad (7)$$

For a white band $C_\ell^{\gamma\gamma} = P_0$ this reduces to $b_\ell^\kappa = P_0 \text{beam2bl}[|K(\theta)|^2]$, i.e. the squared-kernel $|K|^2$ of the optimal-weights note (its noise template, with $n^2 \rightarrow P_0$). The shear power enters only through the *shape* of κ (its width within the band); the response amplitude b_ϕ cancels below.

Transfer. Full sky with $W_1 = 1$ gives the truth response $\delta M_0^t = 2b_\phi \kappa \star \delta$, so $\langle \delta M_0^t \delta \rangle_\ell = 2b_\phi b_\ell^\kappa C_\ell^{\delta\delta}$. NaMaster already removes the cross mode-coupling of the (W_2 mask, mask) weights, so the residual transfer is the field-level window relating the constructed response $\kappa \star (W_1^2 \delta)$ to the truth $\kappa \star \delta$:

$$T_\ell = \frac{\langle \text{decouple}_{W_2 \text{m}} \text{pseudo} C_\ell[(W_2 \text{m}) \kappa \star (W_1^2 \delta), \text{m} \delta] \rangle}{\langle \text{decouple}_1 \text{pseudo} C_\ell[\kappa \star \delta, \delta] \rangle}, \quad (8)$$

independent of b_ϕ and of the normalization of κ .

Density assumption and weight contraction. We treat δ_{src} as an externally *fixed survey window*: it sets the depth and hence the noise and the weights W_1, W_2 , which are therefore deterministic maps (like the mask). The lens density δ — at a different redshift — is the random probe, taken uncorrelated with the window, $\delta_{\text{src}} \perp \delta$. The prediction thus uses only the spectra $C^{\gamma\gamma}$ (via κ) and $C^{\delta\delta}$ (the probe), with δ drawn from its spectrum; it never uses the lens map. Because W_1^2 is fixed and independent of δ , the response $\kappa \star (W_1^2 \delta)$ is linear in δ and the transfer is a functional of $C_\ell^{\delta\delta}$ and the survey maps alone.

Four evaluations.

- (A) *Measured*: from the 400 noise-free signal realizations of the main run (these use the *real*, weakly correlated δ_{src} and δ).
- (B) *Gaussian Monte-Carlo*: a fixed δ_{src} window $\rightarrow$ fixed W_1, W_2 ; draw Gaussian δ from $C_\ell^{\delta\delta}$; propagate $\kappa \star (W_1^2 \delta)$ through the identical NaMaster decoupling and average.
- (C) *Closed form (no Monte-Carlo)*:

$$T_\ell^{(C)} = g_{\text{edge}} \times N_{\text{weight}}, \quad g_{\text{edge}} = \frac{\langle \kappa \star \mathbf{m} \rangle_{\mathbf{m}}}{\langle \mathbf{m} \rangle_{\mathbf{m}}}, \quad N_{\text{weight}} = \frac{\langle W_2 \mathbf{m} W_1^2 \rangle}{\langle W_2 \mathbf{m} \rangle} \quad (9)$$

- (D) *Mode-coupling matrix*: decouple the measured pseudo-spectrum with the NaMaster coupling matrix of an effective window carrying the weight modulation (Eq. (11) below); with the kernel-smoothed window $w_{\text{D}} = W_2 \mathbf{m} (\kappa \star W_1^2)$ the W_1 Jensen factor *and* the edge loss are absorbed as a full matrix, leaving no transfer function at all.

$g_{\text{edge}} \simeq 0.93\text{--}0.94$ is the finite-kernel mask-edge loss (the fraction of the filter footprint inside the mask, set by the filter and C^γ); it sets the overall $\simeq 0.94$ level for the unweighted schemes. N_{weight} is the net inverse-variance Jensen factor: $W_1^2 \propto (1 + \delta_{\text{src}})^2$ upweights low-noise regions, and although the optimal W_2 partly compensates, a residual enhancement survives, $N_{\text{weight}} \approx 1.04\text{--}1.10$ for W1W2 (rising with band) and $= 1$ for **none**/W2. This is precisely the measured W1W2 lift, and it is reproduced by the prediction from the *fixed, uncorrelated* window alone — demonstrating that the lift is a deterministic property of the inverse-variance weighting and the survey depth, not a source–lens density correlation.

Method (D): absorbing W_1 into the mode-coupling matrix. Rather than dividing out a transfer function, the W_1 modulation can be handed to NaMaster itself. If the weight were smooth on the kernel scale the response would commute, $\kappa \star (W_1^2 \delta) \simeq W_1^2 (\kappa \star \delta)$, and the weighted squared field entering the pseudo-spectrum would be

$$(W_2 \mathbf{m}) \delta M_0 \simeq (W_2 \mathbf{m} W_1^2) \times 2b_\phi (\kappa \star \delta), \quad (10)$$

i.e. the *full-sky truth response* observed through the effective multiplicative window $W_2 \mathbf{m} W_1^2$. Decoupling the measured pseudo-spectrum with the mode-coupling matrix computed for the window pair $(\mathbf{m}, w_{\text{D}})$ — in NaMaster, a map-less “dummy” field carrying w_{D} —

$$\hat{C}_b^{(\text{D})} = \sum_{b'} [M^{(\mathbf{m}, w_{\text{D}})}]_{bb'}^{-1} \text{pseudo} C_{b'} [(\mathbf{m}) \delta, (W_2 \mathbf{m}) M_0] \quad (11)$$

then estimates the truth response directly: the inverse-variance Jensen factor (N_{weight} of method (C)) is absorbed into the coupling matrix, as a full ℓ -dependent matrix rather than a scalar. The arbitrary W_1 normalization cancels ($M_0 \propto W_1^2$ and $w_{\text{D}} \propto W_1^2$), and for **none**/W2 ($W_1 = 1$) the construction reduces to the standard decoupling.

The choice of w_{D} requires care, because the survey weight is *not* smooth on the kernel scale (δ_{src} has structure at all ℓ). Writing out the δ -contraction of the pseudo-spectrum,

$$\langle (\mathbf{m} \delta)(x) (W_2 \mathbf{m})(\hat{\mathbf{n}}) [\kappa \star (W_1^2 \delta)](\hat{\mathbf{n}}) \rangle = \int \mathbf{m}(x) (W_2 \mathbf{m})(\hat{\mathbf{n}}) \kappa(\hat{\mathbf{n}} - a) W_1^2(a) \xi_\delta(x - a) da, \quad (12)$$

the weight is probed at the *inner* point a of the kernel convolution: since the probe ξ_δ is long-ranged (the cross lives at low ℓ) while κ is narrow, the multiplicative-window model holds with the *kernel-smoothed* weight,

$$w_D^{\text{smooth}} = W_2 m(\kappa \star W_1^2), \quad (13)$$

(κ normalized to unit integral) rather than with the raw $w_D^{\text{raw}} = W_2 m W_1^2$ of the naive commutation: the raw window over-counts the small-scale weight structure that the kernel filters out of the band. The two differ in their residual calibration. With w_D^{raw} the idealized residual is the kernel-edge factor g_{edge} of Eq. (9) (the smoothing of W_1^2 , which carries the mask, is not represented). With w_D^{smooth} the window itself contains the kernel’s mask-edge dip — $\kappa \star W_1^2$ falls off where the kernel support leaves the footprint, the very effect g_{edge} parameterizes — so the expected residual calibration is *unity*: no transfer function at all.

Three practical points. (i) Both decouplings act linearly on the same measured pseudo-spectrum, so the (D) bandpowers follow from the stored standard-decoupled ones exactly, $\hat{C}^{(D)} = M_D^{-1} M_{\text{std}} \hat{C}^{\text{std}}$ (binned matrices), with no re-measurement; we verified this re-decoupling against a direct (D) decoupling of one realization to relative precision $\lesssim 4 \times 10^{-15}$. (ii) The coupling matrices use the fixed survey window of realization 1 — the same fixed-window treatment as (B)/(C) — while the measured M_0 carries the per-realization W_1 . (The *ensemble-mean* noise map must not be used here: averaging over source-density realizations washes out the modulation, leaving a nearly flat window and a no-op correction.) (iii) For the spin-4 channel the decoupling in principle mixes (E, B); we apply the $E \rightarrow E$ block of the re-decoupling matrix to the stored E bandpowers, having checked that the $E \leftarrow B$ leakage block vanishes identically for these windows.

Figure 13 tests both windows on the signal-only channel (M_0). The raw window *over-corrects*: the residual transfer lands below g_{edge} (medians 0.92 in [1000, 1500] and 0.89 in [500, 1000], against $g_{\text{edge}} \simeq 0.94$), because the coupling matrix charges the estimator for small-scale weight power that the kernel has already filtered out of the band. The kernel-smoothed window behaves as derived: its residual is consistent with *unity* to 2% in [1000, 1500] ($\langle T \rangle = 0.98$) and to 5% in [500, 1000] ($\langle T \rangle = 0.95$) — the W_1 lift *and* the edge loss are both absorbed by the matrix, with no transfer function applied at all. (The two variants differ by a nearly flat factor $\simeq 1/g_{\text{edge}}$: the kernel smoothing of the window is dominated by its mask-edge dip.) The remaining few-percent deficit has the expected sign: the response carries the weight at *two* filtered points, $W_1(a)W_1(b)$, so weight fluctuations above the field bandpass cannot modulate the band at all, whereas $\kappa \star W_1^2$ retains their positive contribution to the local mean square — the matrix still over-corrects slightly. In the wide band [50, 500] the deficit grows to $\sim 15\%$, the regime in which the white-band κ approximation already degrades methods (B) (8%) and (C). The spin-4 channel follows the same pattern within its much larger transfer uncertainty (median smoothed-window residuals 0.87–0.90).

In absolute terms, the truth-residual χ^2 (recipe of Table 2, 400-realization error on the mean) of the transfer-free smoothed-window calibration of **W1W2** M_0 is 96/16, 49/16, 17.5/16 in the three bands, against 57/16, 34/16, 14/16 for the (B)-calibrated and 46/16, 42/16, 18/16 for the (C)-calibrated bandpowers (all $\lesssim 0.15\sigma$ per realization): comparable in the narrow bands, worse in the wide one. Method (D) is thus not more accurate than (B) on this footprint, but it is the only calibration that is simultaneously Monte-Carlo-free, matrix-level (it corrects bandpower-to-bandpower couplings of the weight, not just the mean), and free of any explicit transfer function.

Spin-4. A scalar power modulation does not source $M_{4E} = \text{Re } \xi^2$ at linear order ($\mathbb{E}[\xi^2] = 0$); its signal is the anisotropic (tidal) response, with a different kernel and amplitude. The *geometric* transfer Eq. (9) — mask $\otimes$ filter non-commutation and weight normalization — is, however, common to both spin channels, as confirmed by the near-equal measured M_0 and M_{4E} transfers (Table 2).

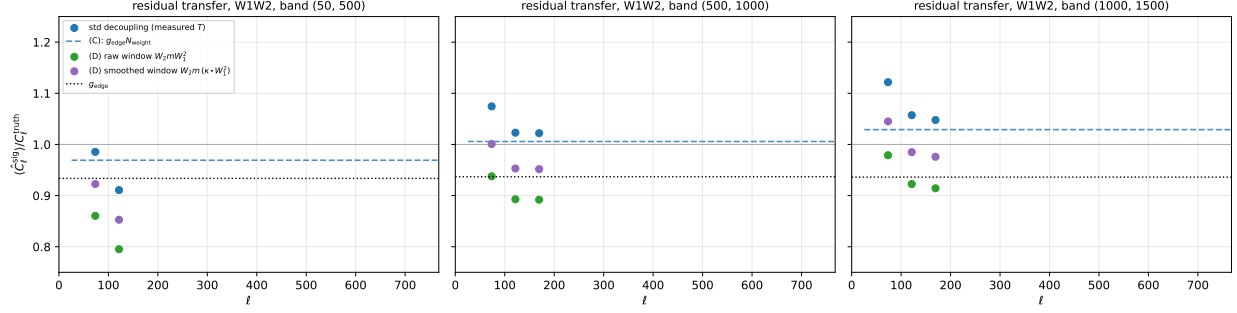


Figure 13: Residual transfer of the **W1W2** scheme under the standard decoupling (blue points: measured; blue dashed: the closed form $g_{edge}N_{weight}$ of Eq. (9)) and under the method-(D) matrix decoupling, with the raw window $W_2mW_1^2$ (green) and the kernel-smoothed window $W_2m(\kappa \star W_1^2)$ of Eq. (13) (purple), for the M_0 channel per filter band, on the signal-bearing bandpowers. The dotted line is g_{edge} , the solid grey line unity. The raw window over-corrects (lands below g_{edge}); the kernel-smoothed window absorbs both the W_1 lift and the edge loss, leaving a residual close to unity in the narrow bands.

We therefore compare the scalar prediction with both measured channels in Fig. 14.

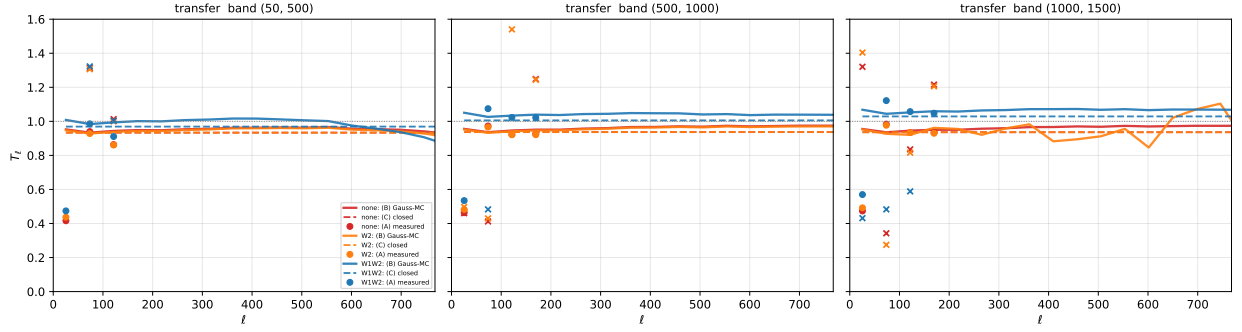


Figure 14: Transfer function by the three methods over the full analysed range $\ell \leq 768$, per scheme (colour) and band. Points: measured (A) for M_0 (circles, signal-bearing bandpowers) and M_{4E} (crosses). Solid lines: Gaussian Monte-Carlo (B); dashed: closed form (C), Eq. (9). The predictions use only the fixed δ_{src} survey window (hence W_1, W_2), the spectra $C^{\gamma\gamma}$ (via κ) and $C^{\delta\delta}$, the mask and the filter — and never the lens map. (B) and (C) agree, and both match the measured transfer to $\sim 1\text{--}3\%$ in the narrow bands $[500, 1000]$ and $[1000, 1500]$ — including the **W1W2** lift to $\simeq 1.0\text{--}1.05$ — and to $\sim 8\%$ in the wide low band $[50, 500]$, where the white-band approximation for κ is least accurate (cf. the optimal-weights note). The lift is thus a deterministic inverse-variance effect, not a source–lens correlation; both spin channels follow the common scalar prediction. (The lowest bandpower of the measured T is noisy: the mode-coupling matrix is ill-conditioned for the largest scales on an $\sim 11\%$ footprint.)